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1 Filling the Gaps: Helping Boilerexams Identify Missing Video Explanations

1.1 Introduction and Problem

Believe it or not, on the **Boilerexams** website, there are exams that don't have videos explaining why the correct answers are valid. For those of you reading this who are spoiled by the video explanations in the Math courses on Boilerexams, go check out the **CS251 exams**. You'll find questions with no videos explaining the reasoning behind the answers. As a CS major, I couldn't stand for this injustice.

Having studied using the CS251 material, I knew the pain of trying to explain to friends **why their answer was correct or wrong**. Without video explanations, we had to piece things together ourselves. That's when I decided to build a tool that would **identify which questions were missing videos**, so the content team could quickly snipe those questions and create videos for them.

1.2 Methodology

Like I said, the problem was that there were questions with no videos explaining why they were being missed. The content team is made up of Purdue students; as a result, they only have so much time. So, if I wanted to help students by giving the content team something useful, I also needed to make sure that I was **identifying the right questions** to be worth the content team's time.

I came up with **three metrics** to prioritize which questions needed videos: 1. **Average Percent Correct** – How often students got the question right. 2. **Total Submissions** – How many times the question was attempted. 3. **Number of Correct Submissions** – How many of those attempts were correct.

1.3 Building the Table

To make this work, I needed to gather data from multiple tables in the database. I combined the **Questions, Exam, Course, and Video** tables to build a complete list of questions without videos.

Here's where it got tricky: some questions were associated with **multiple topics**. For example, a shell sorting question might be under both **Sorting** and **Shell Sorting**. To clean this up, I grouped all the duplicate questions together so that **each question appeared only once**, but with all its associated topics listed.

With the table ready, I moved on to making the tool useful for the content team.

1.4 Building the Interface

The table was good, but I wanted to make it **easy for the content team to use**. So, I built an interface using **Jupyter Notebook's ipywidgets**. The interface lets the team:
- **Select a course** they're interested in.
- **Sort questions** by any of the three metrics to focus on the most important questions.

1.5 The Final Tool

The tool is a simple but valuable table that gives the content team everything they need to decide which questions should get video explanations. No more guessing—just data to help them focus on the most-missed and high-impact questions.

1.6 What's Next?

As I'm writing this, I've been thinking of a few ways to make the tool even better:

1. **Automated Data Updates** – A script that automatically refreshes the table with new data so the team always has the latest info.

2. **Integration with the Course Creation Platform** – Embedding the tool directly into the course creation site so the content team doesn't have to load up the notebook.

3. **Real-Time Insights on Most-Missed Questions** – Adding a feature that tracks a **running average of the most-missed questions** leading up to an exam. This would be super helpful for:
    - For courses with fewer exams or that are newer, the tool helps prioritize missing videos for upcoming exams based on recent student struggles.
    - For courses with many exams, it focuses on aligning video creation with current student needs by analyzing recent data.

1.7 Why This Matters?

This tool isn't just a cool project. It's something that can **augment the content team's time and effort** by giving them the data they need to make smart decisions about where to focus their work. More importantly, it makes sure that future students won't have to struggle through tricky questions without video explanations—something I wish I had when I was taking CS251.

By helping the content team target the right questions, this tool makes sure that **students get the explanations they need** to succeed.

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